Retail Strategy and Analytics

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DATA PREPARATION

1. Load the required libraries

```
library(data.table)
library(ggplot2)
library(dplyr)
library(skimr)
library(readxl)
library(readr)
library(tidyr)
```

2. Set working directory

```
setwd("C:/Users/LENOVO/Documents/KhanhMai/1. Personal/Projects/Forage/04 Quantium - Data Analytics")
```

3. Import data

```
customerD <- read_csv("QVI_purchase_behaviour.csv")
transactionD <- read_excel("QVI_transaction_data.xlsx")
```

DATA EXPLORATION

1. Examining transaction data

```
head(transactionD)
```

```
## # A tibble: 6 x 8
##    DATE STORE_NBR LYLTY_CARD_NBR TXN_ID PROD_NBR PROD_NAME    PROD_QTY TOT_SALES
##   <dbl>     <dbl>          <dbl>  <dbl>    <dbl> <chr>           <dbl>     <dbl>
## 1 43390         1           1000      1        5 Natural Chi~        2       6
## 2 43599         1           1307    348       66 CCs Nacho C~        3       6.3
## 3 43605         1           1343    383       61 Smiths Crin~        2       2.9
## 4 43329         2           2373    974       69 Smiths Chip~        5      15
## 5 43330         2           2426   1038      108 Kettle Tort~        3      13.8
## 6 43604         4           4074   2982       57 Old El Paso~        1       5.1
```

```r
str(transactionD)
```

```
## tibble [264,836 x 8] (S3: tbl_df/tbl/data.frame)
##  $ DATE          : num [1:264836] 43390 43599 43605 43329 43330 ...
##  $ STORE_NBR     : num [1:264836] 1 1 1 2 2 4 4 4 5 7 ...
##  $ LYLTY_CARD_NBR: num [1:264836] 1000 1307 1343 2373 2426 ...
##  $ TXN_ID        : num [1:264836] 1 348 383 974 1038 ...
##  $ PROD_NBR      : num [1:264836] 5 66 61 69 108 57 16 24 42 52 ...
##  $ PROD_NAME     : chr [1:264836] "Natural Chip        Compny SeaSalt175g" "CCs Nacho Cheese    175g
##  $ PROD_QTY      : num [1:264836] 2 3 2 5 3 1 1 1 1 2 ...
##  $ TOT_SALES     : num [1:264836] 6 6.3 2.9 15 13.8 5.1 5.7 3.6 3.9 7.2 ...
```

Convert DATE column to a date format

```r
transactionD$DATE <- as.Date(as.numeric(transactionD$DATE), origin = "1899-12-30")
```

Examine PROD_NAME

```r
summary(transactionD$PROD_NAME)
```

```
##    Length     Class      Mode
##    264836 character character
```

Now check if there are all chips as we are only interested in chips-related products Summarise individual words in PROD_NAME

```r
productWords <- data.table(
  words = unlist(strsplit(unique(transactionD$PROD_NAME), " ")))
```

The only interest is in words telling if the product is chips or not Let's remove digits, special characters from these individual words

```r
productWords_clean<- productWords[!grepl("[^A-Za-z]",words)]
```

And sort these distinct words by frequency from highest to lowest

```r
wordfreq <- productWords_clean[, .N, by = words][order(-N)]
```

There are salsa products in the dataset but chip products are the sole interest so let's remove salsa products, to avoid unnecessary data loss

```r
transactionD_2 <- transactionD[!grepl("salsa",transactionD$PROD_NAME, ignore.case = TRUE),]
```

Summarise the data to check for nulls and possible outliers

Let's check for nulls

```r
colSums(is.na(transactionD_2))
```

```
##           DATE      STORE_NBR LYLTY_CARD_NBR         TXN_ID       PROD_NBR
##              0              0              0              0              0
##      PROD_NAME       PROD_QTY      TOT_SALES
##              0              0              0
```

There are no null values in the column

```r
#### Summarise numeric columns for outliers
summary(transactionD_2[, sapply(transactionD_2, is.numeric)])
```

```
##    STORE_NBR     LYLTY_CARD_NBR        TXN_ID           PROD_NBR
##  Min.   :  1.0   Min.   :   1000   Min.   :      1   Min.   :  1.00
##  1st Qu.: 70.0   1st Qu.:  70015   1st Qu.:  67569   1st Qu.: 26.00
##  Median :130.0   Median : 130367   Median : 135183   Median : 53.00
##  Mean   :135.1   Mean   : 135531   Mean   : 135131   Mean   : 56.35
##  3rd Qu.:203.0   3rd Qu.: 203084   3rd Qu.: 202654   3rd Qu.: 87.00
##  Max.   :272.0   Max.   :2373711   Max.   :2415841   Max.   :114.00
##     PROD_QTY         TOT_SALES
##  Min.   :  1.000   Min.   :  1.700
##  1st Qu.:  2.000   1st Qu.:  5.800
##  Median :  2.000   Median :  7.400
##  Mean   :  1.908   Mean   :  7.321
##  3rd Qu.:  2.000   3rd Qu.:  8.800
##  Max.   :200.000   Max.   :650.000
```

Product quantity column seems to have an outlier which is the max value, need further investigation

Filter the dataset to find the outlier

```r
transactionD_filter <- transactionD_2 %>% filter(PROD_QTY==200)
```

There are two transactions of 200 chips packets bought by the same customer, which needs further investigation on the customer to see if they made any other transaction

```r
transactionD_filter1 <- transactionD_2 %>% filter(LYLTY_CARD_NBR==226000)
```

This customer only had two purchases throughout the year, indicating non-retail customer. This might just a customer buying bulk or for commercial purposes, which should not be included in this retail customer analysis to avoid distortion. This loyalty card number will be removed from further analysis.

```r
#### Filter out the outliers
transactionD_3 <- transactionD_2%>%filter(!LYLTY_CARD_NBR == 226000)
```

Re-examine transaction data

```r
summary(transactionD_3)
```

```
##       DATE              STORE_NBR     LYLTY_CARD_NBR        TXN_ID
##  Min.   :2018-07-01   Min.   :  1.0   Min.   :   1000   Min.   :      1
##  1st Qu.:2018-09-30   1st Qu.: 70.0   1st Qu.:  70015   1st Qu.:  67569
##  Median :2018-12-30   Median :130.0   Median : 130367   Median : 135182
##  Mean   :2018-12-30   Mean   :135.1   Mean   : 135530   Mean   : 135130
##  3rd Qu.:2019-03-31   3rd Qu.:203.0   3rd Qu.: 203083   3rd Qu.: 202652
##  Max.   :2019-06-30   Max.   :272.0   Max.   :2373711   Max.   :2415841
##     PROD_NBR       PROD_NAME            PROD_QTY       TOT_SALES
##  Min.   :  1.00   Length:246740      Min.   :1.000   Min.   : 1.700
##  1st Qu.: 26.00   Class :character   1st Qu.:2.000   1st Qu.: 5.800
##  Median : 53.00   Mode  :character   Median :2.000   Median : 7.400
##  Mean   : 56.35                      Mean   :1.906   Mean   : 7.316
##  3rd Qu.: 87.00                      3rd Qu.:2.000   3rd Qu.: 8.800
##  Max.   :114.00                      Max.   :5.000   Max.   :29.500
```

Date column has only 364 unique values, meaning there is a missing date that need to be found

Create a sequence of dates from 1 Jul 2018 to 30 Jun 2019

```r
#### Create a sequence of dates and join this the count of transactions by date to find missing date
yeardates <- data.table(DATE = seq(
  from = as.Date("2018-07-01"),
  to = as.Date("2019-06-30"),
  by ="day"))
```

```r
#### Join the two DATE columns to see the missing one
missing_date <- anti_join(yeardates,transactionD_3,by = "DATE" )
```

The missing date is 25 Dec 2018, Christmas day

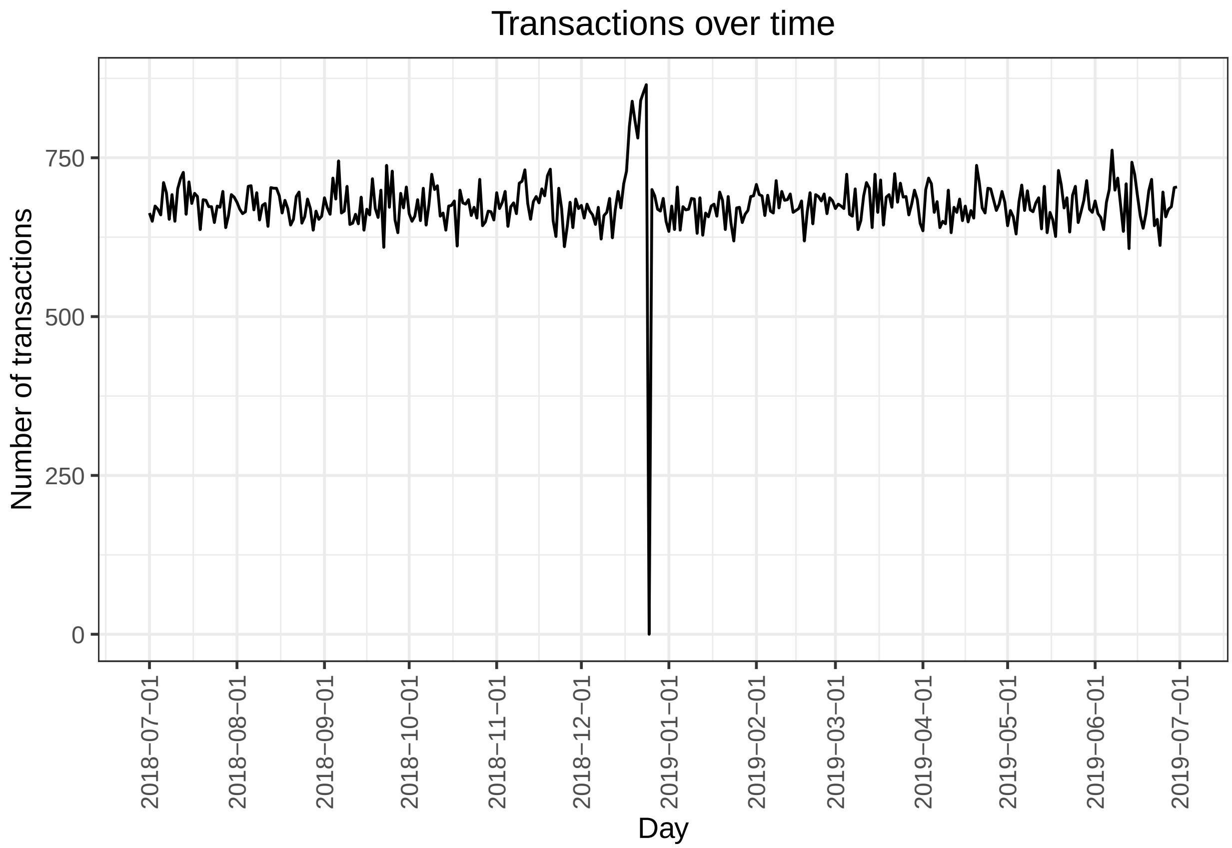
Plot transactions over time

Include missing date with 0 transaction count

```r
transaction_bydate <- transactionD_3%>%summarise(N=n(),.by=DATE)
transaction_bydate <- full_join(yeardates,transaction_bydate,by = "DATE")
transaction_bydate[is.na(N), N := 0]
```

Create a plot showing the number of transactions over time

```r
ggplot(transaction_bydate, aes(x = DATE, y = N)) +
  geom_line() +
  labs(x = "Day", y = "Number of transactions", title = "Transactions over time") +
  scale_x_date(breaks = "1 month") +
  theme_bw()+
  theme(plot.title = element_text(hjust = 0.5),
        axis.text.x = element_text(angle = 90, vjust = 0.5))
```



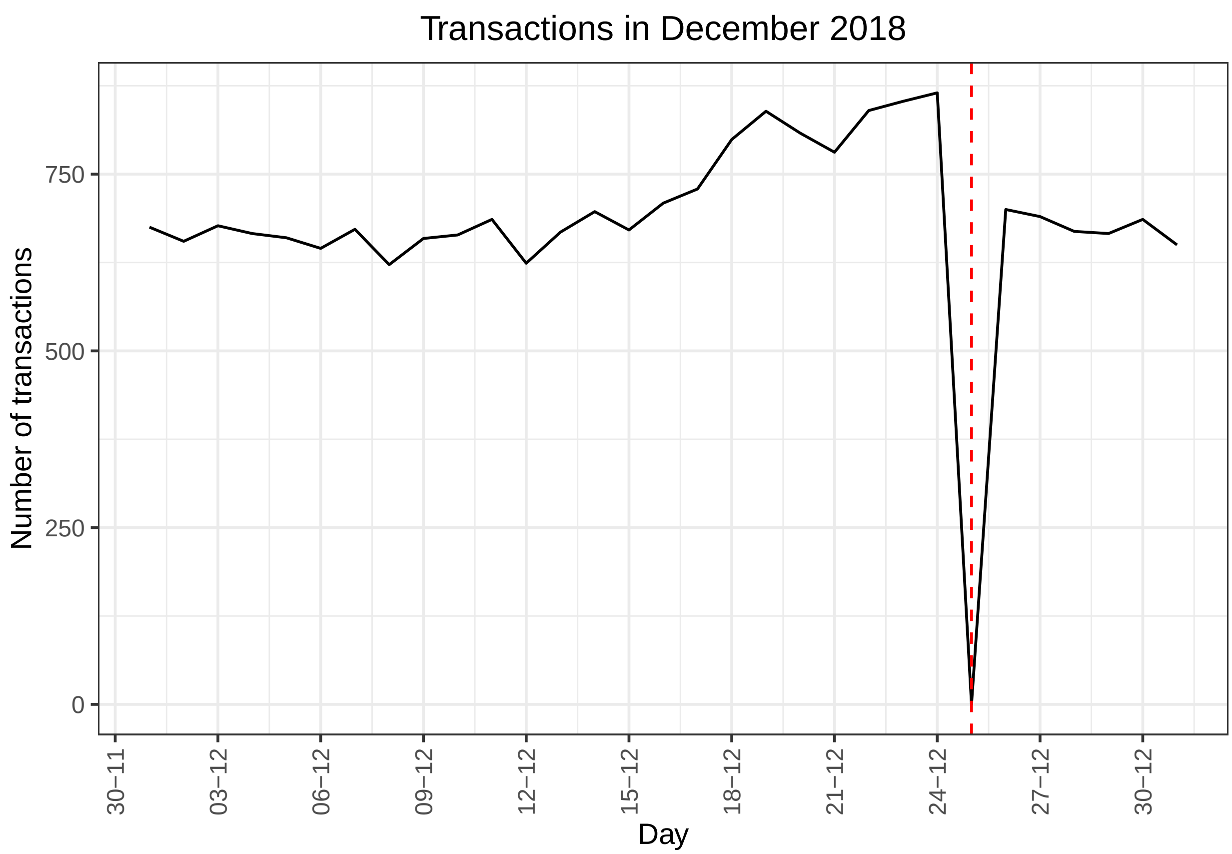
There is an increase in purchases in December and a break in late December

Filter to December and look at individual days

```
transaction_Dec <- transaction_bydate %>% filter(DATE >= as.Date("2018-12-01") & DATE <= as.Date("2018-
```

Let's look at the plot showing the number of transactions over December 2018

```
ggplot(transaction_Dec, aes(x = DATE, y = N)) +
  geom_line() +
  labs(x = "Day", y = "Number of transactions", title = "Transactions in December 2018") +
  scale_x_date(breaks = "3 days", date_labels = "%d-%m") +
  theme_bw()+
  theme(plot.title = element_text(hjust = 0.5),
        axis.text.x = element_text(angle = 90, vjust = 0.5))+
  geom_vline(xintercept = as.Date("2018-12-25"), linetype = "dashed", color = "red")
```



The increase in sales occurs in the lead-up to Christmas and that there are zero sales on Christmas day itself. This is due to shops being closed on Christmas day. Now the data no longer has outliers, move on to creating other features such as brand of chips or pack size from PROD_NAME.

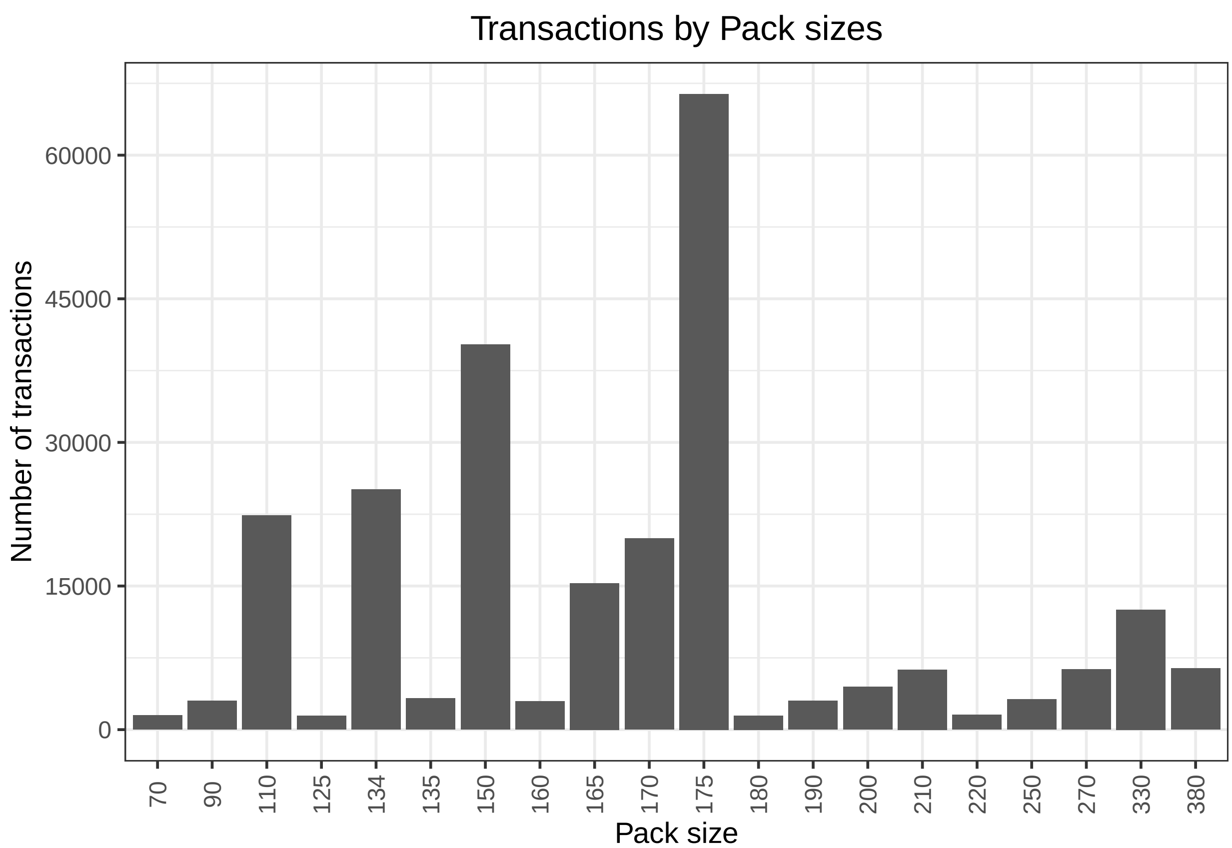
Pack size

```
#### Add a pack size column from PROD_NAME by taking the digits
transactionD_3<-transactionD_3%>% mutate(PACK_SIZE = parse_number(PROD_NAME))
```

```
#### Check the column if it looks sensible
transaction_bysize<-transactionD_3%>%count(PACK_SIZE, sort = TRUE)
```

The largest size is 380g and the smallest is 70g, seems sensible

```
#### Plot a histogram as PACK_SIZE is a categorical variable although it is numeric
ggplot(transactionD_3,aes(factor(PACK_SIZE)))+
  geom_bar()+
  labs(x = "Pack size", y = "Number of transactions", title = "Transactions by Pack sizes")+
  theme_bw()+
  theme(plot.title = element_text(hjust = 0.5),
        axis.text.x = element_text(angle = 90, vjust = 0.5))+
  scale_y_continuous(breaks = seq(0, max(table(transactionD_3$PACK_SIZE)) + 15000, by = 15000))
```



Brands

```
#### Add a brand name column from PROD_NAME by extracting the first word
transactionD_3<-transactionD_3%>% mutate(BRAND=sub(" .*", "", transactionD_3$PROD_NAME))
```

```
#### Check brand names
unique(transactionD_3$BRAND)
```

```
##  [1] "Natural"    "CCs"        "Smiths"     "Kettle"     "Grain"
##  [6] "Doritos"    "Twisties"   "WW"         "Thins"      "Burger"
## [11] "NCC"        "Cheezels"   "Infzns"     "Red"        "Pringles"
## [16] "Dorito"     "Infuzions"  "Smith"      "GrnWves"    "Tyrrells"
## [21] "Cobs"       "French"     "RRD"        "Tostitos"   "Cheetos"
## [26] "Woolworths" "Snbts"      "Sunbites"
```

Some of the brand names look like they are of the same brands. Let's combine these together

```
#### Clean brand names by combing same brands
transactionD_3 <- transactionD_3 %>%
  mutate(BRAND = case_when(
      BRAND == "Red" ~ "RRD",
      BRAND == "Dorito" ~ "Doritos",
      BRAND == "Smith" ~ "Smiths",
```

```r
      BRAND == "Infzns" ~ "Infuzions",
      BRAND == "Snbts" ~ "Sunbites",
      BRAND == "WW" ~ "Woolworths",
      TRUE ~ BRAND))
unique(transactionD_3$BRAND)
```

```
##  [1] "Natural"    "CCs"        "Smiths"     "Kettle"     "Grain"
##  [6] "Doritos"    "Twisties"   "Woolworths" "Thins"      "Burger"
## [11] "NCC"        "Cheezels"   "Infuzions"  "RRD"        "Pringles"
## [16] "GrnWves"    "Tyrrells"   "Cobs"       "French"     "Tostitos"
## [21] "Cheetos"    "Sunbites"
```

```r
#### Check the brands again with its distribution
transaction_bybrand<-transactionD_3 %>%
  count(BRAND, sort = TRUE)
```

This looks reasonable

2. Examining customer data

```r
head(customerD)
```

```
## # A tibble: 6 x 3
##   LYLTY_CARD_NBR LIFESTAGE              PREMIUM_CUSTOMER
##            <dbl> <chr>                  <chr>
## 1           1000 YOUNG SINGLES/COUPLES  Premium
## 2           1002 YOUNG SINGLES/COUPLES  Mainstream
## 3           1003 YOUNG FAMILIES         Budget
## 4           1004 OLDER SINGLES/COUPLES  Mainstream
## 5           1005 MIDAGE SINGLES/COUPLES Mainstream
## 6           1007 YOUNG SINGLES/COUPLES  Budget
```

```r
summary(customerD)
```

```
##  LYLTY_CARD_NBR     LIFESTAGE         PREMIUM_CUSTOMER
##  Min.   :   1000   Length:72637       Length:72637
##  1st Qu.:  66202   Class :character   Class :character
##  Median : 134040   Mode  :character   Mode  :character
##  Mean   : 136186
##  3rd Qu.: 203375
##  Max.   :2373711
```

Take a look at the distributions

```r
table(customerD$LIFESTAGE)
```

```
##
## MIDAGE SINGLES/COUPLES           NEW FAMILIES         OLDER FAMILIES
##                   7275                   2549                   9780
```

```
##  OLDER SINGLES/COUPLES               RETIREES        YOUNG FAMILIES
##                  14609                  14805                  9178
##  YOUNG SINGLES/COUPLES
##                  14441
```

```
table(customerD$PREMIUM_CUSTOMER)
```

```
##
##     Budget Mainstream    Premium
##      24470      29245      18922
```

Both variables are fine, no missing values or outliers

3. Merging transaction and customer tables

```
data <- left_join(transactionD_3,customerD,"LYLTY_CARD_NBR")
```

When join two tables, there might be missing customer details

```
colSums(is.na(data))
```

```
##             DATE        STORE_NBR   LYLTY_CARD_NBR           TXN_ID
##                0                0                0                0
##         PROD_NBR        PROD_NAME         PROD_QTY        TOT_SALES
##                0                0                0                0
##        PACK_SIZE            BRAND        LIFESTAGE PREMIUM_CUSTOMER
##                0                0                0                0
```

There are no nulls, so all customers in transaction data has been included in the customer data.

DATA ANALYSIS

Data analysis on customer segments

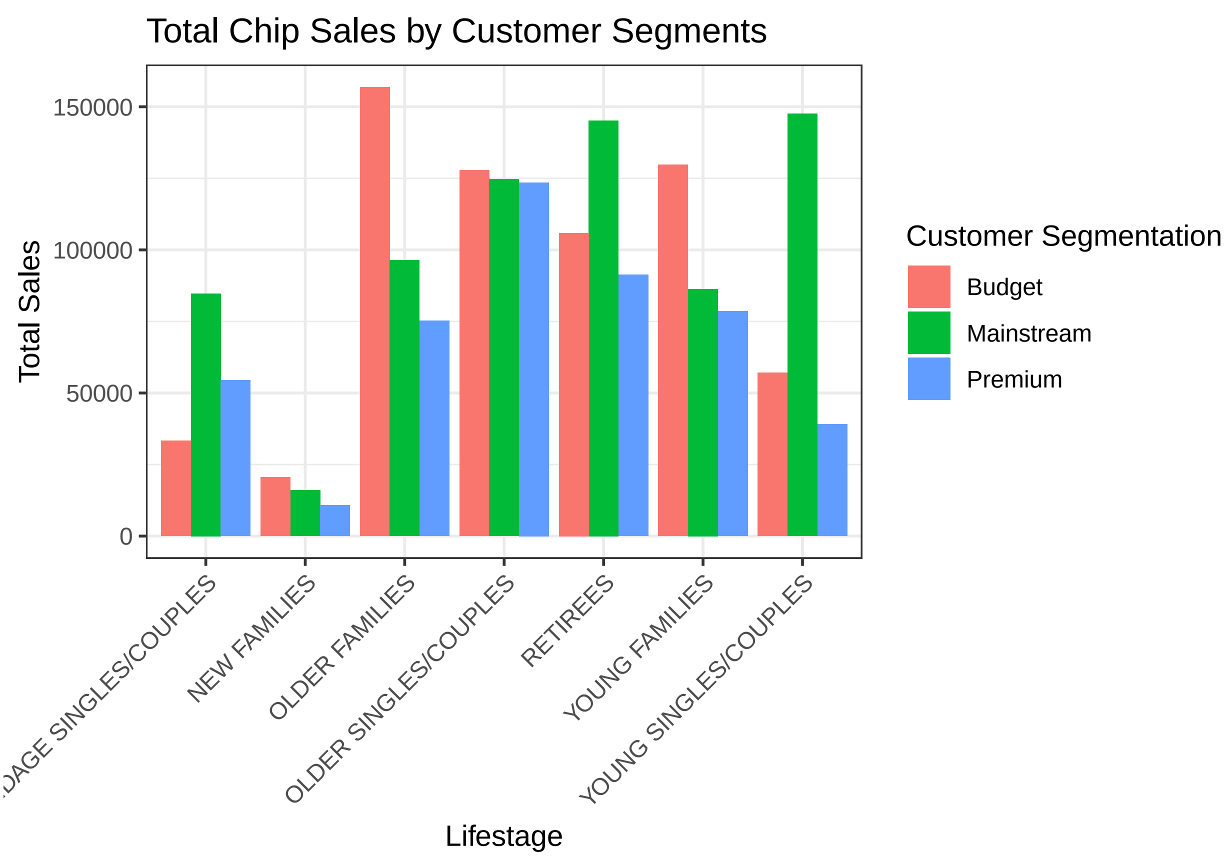
1. Total sales by PREMIUM_CUSTOMER and LIFESTAGE

Summarise sales by PREMIUM_CUSTOMER and LIFESTAGE

```
sales_bysegments <- data %>%
  summarise(sales = sum(TOT_SALES),.by = c(PREMIUM_CUSTOMER, LIFESTAGE)) %>%
  arrange(desc(sales))
```

Create a plot indicating the sales by PREMIUM_CUSTOMER and LIFESTAGE

```r
ggplot(sales_bysegments,
       aes(x = LIFESTAGE, y = sales, fill = PREMIUM_CUSTOMER)) +
  geom_col(position = "dodge") +
  labs(title="Total Chip Sales by Customer Segments",
    x="Lifestage",
    y="Total Sales",
    fill="Customer Segmentation") +
  theme_bw() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Sales comes largely from Budget - older families, Mainstream - young singles/couples, and Mainstream - retirees., Let's see if the higher sales are due to there being more customers who buy chips.

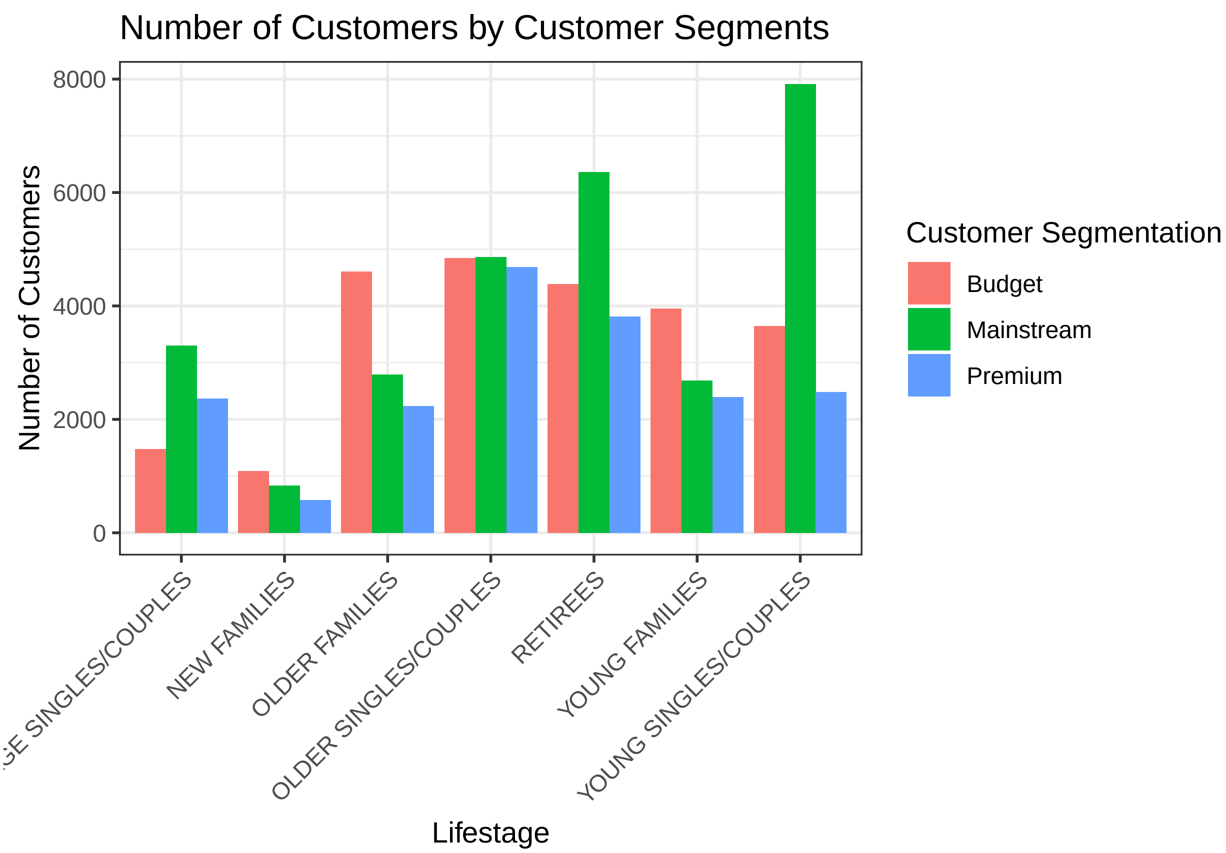
2. Number of customer by PREMIUM_CUSTOMER and LIFESTAGE

Summarise number of customers by PREMIUM_CUSTOMER and LIFESTAGE

```r
count_customer <- data %>%
  summarise(count = n_distinct(LYLTY_CARD_NBR),.by = c(PREMIUM_CUSTOMER, LIFESTAGE)) %>%
  arrange(desc(count))
```

Create a plot indicating the number of customer by PREMIUM_CUSTOMER and LIFESTAGE

```
ggplot(count_customer,
       aes(x = LIFESTAGE, y = count, fill = PREMIUM_CUSTOMER)) +
  geom_col(position = "dodge") +
  labs(title="Number of Customers by Customer Segments",
       x="Lifestage",
       y="Number of Customers",
       fill="Customer Segmentation") +
  theme_bw() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



The highest number of customers who buy chips are Mainstream - young singles/couples and Mainstream - retirees. This contributes to more sales in these segments while this number is not the major driver for Budget - older families segment.

Let's have a deeper look at this as higher sales can derive from other elements like more units of chips being purchased per customers

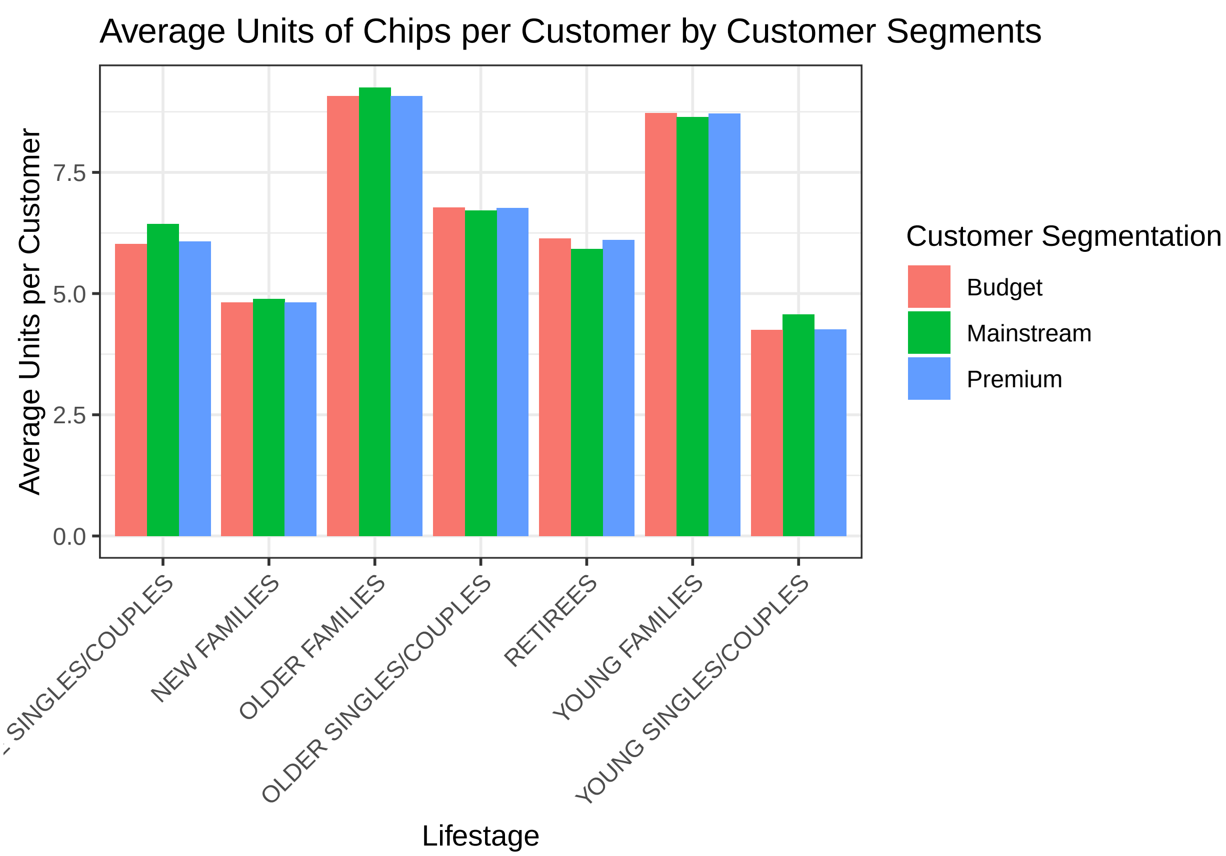
3. Average number of chip units per customer

Summarise average chip units purchased per customer

```
avg_percustomer <- data %>%
  summarise(avg_unit = sum(PROD_QTY)/n_distinct(LYLTY_CARD_NBR),.by = c(PREMIUM_CUSTOMER, LIFESTAGE)) %
  arrange(desc(avg_unit))
```

Create a plot indicating the average number of chip units per customer by PREMIUM_CUSTOMER and LIFESTAGE

```
ggplot(avg_percustomer,
       aes(x = LIFESTAGE, y = avg_unit, fill = PREMIUM_CUSTOMER)) +
  geom_col(position = "dodge") +
  labs(title="Average Units of Chips per Customer by Customer Segments",
       x="Lifestage",
       y="Average Units per Customer",
       fill="Customer Segmentation") +
  theme_bw() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Older families and young families buy more chips per individual in general Let's also investigate the average price per unit chips bought for each customer segment as this is also a driver of total sales.

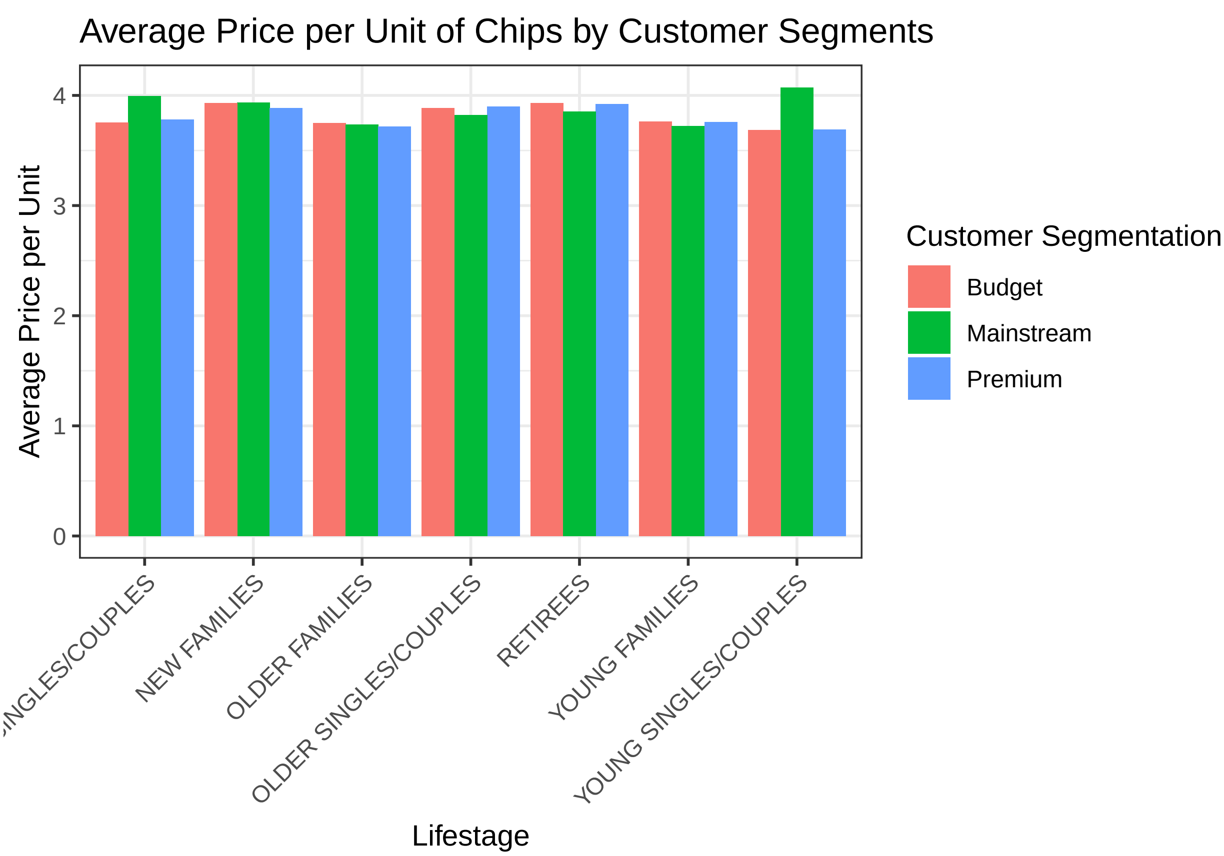
4. Average price per unit by PREMIUM_CUSTOMER and LIFESTAGE

Summarise unit price by PREMIUM_CUSTOMER and LIFESTAGE

```
avgprice_bysegments <- data %>%
  summarise(avg_price = sum(TOT_SALES)/sum(PROD_QTY),.by = c(PREMIUM_CUSTOMER, LIFESTAGE)) %>%
  arrange(desc(avg_price))
```

Create a plot indicating the average price per chip unit by PREMIUM_CUSTOMER and LIFESTAGE

```
ggplot(avgprice_bysegments,
       aes(x = LIFESTAGE, y = avg_price, fill = PREMIUM_CUSTOMER)) +
  geom_col(position = "dodge") +
  labs(title="Average Price per Unit of Chips by Customer Segments",
       x="Lifestage",
       y="Average Price per Unit",
       fill="Customer Segmentation") +
  theme_bw() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Mainstream - midage and young singles and couples are more willing to pay more per packet of chips compared to their budget and premium counterparts. This may be due to premium consumers are more likely to purchase healthy snacks, and when they do purchase chips, it's primarily for amusement rather than personal consumption. This is also supported by there being fewer premium midage and young singles and couples buying chips compared to their mainstream counterparts.

As the difference in average price per unit isn't large, we can check if this difference is statistically significant.

5. Perform an independent t-test between mainstream vs premium and budget mid-age and young singles/couples

Prepare the data for the t-test

```
t_test_data <- data %>%
  filter(LIFESTAGE %in% c("MIDAGE SINGLES/COUPLES", "YOUNG SINGLES/COUPLES")) %>%
  mutate(PRICE_PER_UNIT = TOT_SALES/PROD_QTY)
```

Perfom t-test between mainstream and premium/budget segments

```
#### t-test between mainstream vs premium
t.test(PRICE_PER_UNIT ~ PREMIUM_CUSTOMER,
  data = filter(t_test_data, PREMIUM_CUSTOMER %in% c("Mainstream", "Premium")))
```

```
##
##  Welch Two Sample t-test
##
## data:  PRICE_PER_UNIT by PREMIUM_CUSTOMER
## t = 28.338, df = 23872, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group Mainstream and group Premium is not e
## 95 percent confidence interval:
##  0.2930718 0.3366263
## sample estimates:
## mean in group Mainstream    mean in group Premium
##                 4.039786                 3.724937
```

```
#### t-test between mainstream vs budget
t.test(PRICE_PER_UNIT ~ PREMIUM_CUSTOMER,
  data = filter(t_test_data, PREMIUM_CUSTOMER %in% c("Mainstream", "Budget")))
```

```
##
##  Welch Two Sample t-test
##
## data:  PRICE_PER_UNIT by PREMIUM_CUSTOMER
## t = -31.671, df = 23526, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group Budget and group Mainstream is not equ
## 95 percent confidence interval:
##  -0.3738041 -0.3302324
## sample estimates:
##     mean in group Budget mean in group Mainstream
##                 3.687768                 4.039786
```

Independent t-test results have the p-value < 0.05 at 95% confidence interval for both tests. This indicates the differences are statistically significant. In other words, the unit price for mainstream young and mid-age singles/couples are statistically significantly higher than that for budget or premium young and mid-age singles/couples. It suggests that increased willingness to pay among mainstream young and mid-age customers contributes to higher overall sales.

6. Further investigation on specific customer segments

Some interesting insights have been found from the analysis above. Now let's dive deeper into the target customer segments that contribute the most to chip sales to retain them or further increase sales. Let's look at Mainstream - young singles/couples.

Add a column to identify target customers and the rest

```
data <- data %>%
  mutate(TARGET_SEGMENT =
           if_else(PREMIUM_CUSTOMER == "Mainstream" &
                     LIFESTAGE == "YOUNG SINGLES/COUPLES",
                     "Target","Other"))
```

As the price per chip units of these groups are generally higher than the others, they might tend to buy larger packs of chips. In this part, affinity analysis is used to compare purchase behaviours and preferences between target segments and the rest of customer base.

Pack size affinity analysis

This would examine whether the target segment - Mainstream young singles/couples are more or less likely to buy certain pack sizes compared to the other segments. Let's see the frequency of each pack size purchased and how likely a given size is purchased between each segment.

```
pack_affinity <- data %>%
  count(TARGET_SEGMENT, PACK_SIZE) %>%
  group_by(TARGET_SEGMENT) %>%
  mutate(prop = n/sum(n))
```

```
#### Now compare pack size selection association between target and non-target customers
pack_compare <- pack_affinity %>%
  select(-n)%>%
  pivot_wider(names_from = TARGET_SEGMENT, values_from = prop)%>%
  mutate(lift = Target/Other)%>%
  arrange(desc(lift))
```

Sizes that have lift affinity values more than 1 are mostly larger ones, indicating target customers are significantly more likely to buy bigger packs compared to other customers

The target customers might also focus on particular favourite brands of chips. Let's dive into this.

Brands affinity analysis

This would identify if the segment of interest have stronger preference for brands than the rest.

```
brand_affinity <- data%>%
  count(TARGET_SEGMENT,BRAND)%>%
  group_by(TARGET_SEGMENT)%>%
  mutate(prop = n/sum(n))
```

```
#### Now compare the choice of brand between target and non-target customers
brand_compare <- brand_affinity %>%
  select(-n)%>%
  pivot_wider(names_from = TARGET_SEGMENT, values_from = prop)%>%
  mutate(lift = Target/Other)%>%
  arrange(desc(lift))
```

Lift affinity values for brands over 1 show a brand-driven purchasing behaviour within target segment, who seems to be more attracted to well-known, premium or flavoured brands such as Tyrrells, Twisties, Doritos,… than the other customers.

Insights

This mainstream young customers have a stronger preference for taste and brands over price and tends to purchase larger packs of chips, suggesting brand-focused promotions on visibility for these labels or on bundled deals, family or party packs focused for this segment.